Opening Up Official Statistics in R with pxweb

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Abstract The **pxweb** package provides an interface from R to statistical databases published through the PxWeb API. PxWeb is used by national, regional, and international statistical organizations to disseminate official statistics in multidimensional tables. The package supports discovery of API resources, construction and validation of table queries, retrieval and parsing of data and metadata, conversion to standard R data structures, extraction of comments, and citation of retrieved data. This article introduces the motivation and design of **pxweb**, outlines the support for both PxWeb API version 1 and version 2, and provides a reproducible case study using official statistics.

1 Introduction

Open data science workflows rely heavily on algorithmic tools for data retrieval. Automating major parts of the data workflow, such as finding, accessing, integrating, citing, and reporting data, helps users spend more time on statistical analysis and interpretation. By automating parts of the data workflow we can help users to focus their time on analysis, rather than finding, accessing, and building references for data. It also improves reproducible research (Gandrud, 2013; Boettiger et al., 2015) by making data acquisition both explicit and repeatable.

Official statistics are a particularly important type of public data (United Nations General Assembly, 2014). National statistical institutes (NSI) and international agencies publish large amounts of data in a vast number of areas, such as demography, economics, health, infrastructure, climate, to name a few (United Nations General Assembly, 2014; World Bank, 2026). Such data are often multidimensional, geographically detailed, and updated over long time periods. During the last decade this data has become increasingly available through open web-based application programming interfaces (API). Opening statistical resources through public web APIs is, however, only the first step to enable use of statistical data. Users also need software that can bridge the gap between online statistical databases and analysis in statistical programming environments such as R.

There are many dedicated software packages that simplify and automate analysis workflows. They take into account variations in raw data formats, access details, metadata conventions, and typical use cases so that users can avoid repetitive programming tasks and reduce the risk of processing errors (Groves and Lyberg, 2010). In R, several packages support programmatic data retrieval in different ways, from specific sources (Lahti et al., 2017; Eugster and Schlesinger, 2013; Arel-Bundock, 2026), broader compiled sources such as economic and financial time-series data (Reinhart, 2024), generic data sources (Szöcs et al., 2020; Caro and Biecek, 2017), or specific statistical standards, such as the SDMX data standard (Blondel et al., 2025).

The **pxweb** package (Magnusson et al., 2026) fills a complementary role: it provides a general R interface to databases exposed through the PxWeb application programming interface (API) (Statistics Sweden and Statistics Norway, 2026a). The package can navigate PxWeb resources, fetch metadata, construct and validate queries, retrieve data, convert responses into standard R data structures. The package source submitted with this manuscript corresponds to a forthcoming version 1.0.0 release; it includes 46 API catalogue entries and supports both PxWeb API version 1 and 2.

Data citation is another motivation for the package. Data reuse should document source organizations, accessed resources, version information, and retrieval times where possible (Parsons et al., 2019). The **pxweb** package helps standardize and streamline this process for PxWeb data by collecting source information from the API and data objects and by

encouraging users to cite both the retrieved data and the software used to retrieve them for full reproducibility.

The **pxweb** package was first released on CRAN in 2014 and was substantially upgraded in 2018 and 2026. Recent development in 2026 has added support for the PxWeb API version 2, including table search, `JSON-stat2` responses, v2 metadata, codelists, value sets, aggregation helpers, and observation status comments. The current package is part of the rOpenGov open data science project (Lahti et al., 2013).

1.1 PxWeb and PC-Axis

PxWeb is a web application and API for disseminating PC-Axis-type statistical data (Statistics Sweden and Statistics Norway, 2026a). In turn, PC-Axis data are data files typically organized as multidimensional tables with dimensions such as region, time, sex, age, indicator, or content. The PX software family (including both PC-Axis and PxWeb) is used by a large number of NSIs, e.g. Statistics Sweden, Statistics Finland, and Statistics Norway (Statistics Sweden, 2026; Statistics Finland, 2026; Statistics Norway, 2026) to name a few. PxWeb exposes data and metadata through hypertext transfer protocol (HTTP) endpoints that can be accessed programmatically.

The PX file format stores metadata and data in a single file. The specification lists 1990 as the introduction year for `CONTENTS`, `DATA`, `MATRIX`, `STUB`, and `CODES` (Likidis, 2013, Section 3.1). Statistics Sweden describes PX as a standard format for statistics files used by many statistical offices and as the basis for tools such as PxWin, PxWeb, and PxEdit (Statistics Sweden, 2026). PxWeb extends this file-oriented tradition to online dissemination by publishing statistical tables through web interfaces (Statistics Sweden and Statistics Norway, 2026a).

Recently, PxWeb API version 2 emerged as a joint Statistics Sweden and Statistics Norway project initiated in January 2024 because the older PxWeb version 1 interface had become difficult to develop further and user needs were changing. Public test versions appeared in 2024, followed by a PxWebApi v2 release in September 2025 and a PxWeb 2 release in October 2025. The new API modernizes the infrastructure through GET URL support, JSON-stat2 output, improved filtering, language handling, and richer metadata access (Statistics Sweden and Statistics Norway, 2026c,b,d).

From the perspective of an R user, PxWeb APIs have three important parts. First, a table query must be constructed from table metadata. Second, the data response has to be retrieved and converted into an analysis-ready format in R. Third, the downloaded data should be easy to cite to facilitate provenance and reproducibility. The **pxweb** package handles all three parts for the R user.

The **pxweb** package is solely designed for PxWeb APIs, while packages such as **pxR** focus on parsing legacy PC-Axis files (Bellosta et al., 2025). This distinction is important because API access requires metadata discovery, query construction, request validation, and response parsing rather than only local file import.

Since PxWeb installations are configured by individual providers, such as NSIs, API version support and request limits are endpoint-level properties rather than package-level constants. The **pxweb** package therefore collects available provider configuration metadata, including version 1 and 2 `config` responses, and uses this information when validating requests, making calls, and splitting large downloads.

1.2 Related software

The R ecosystem contains several packages for accessing official statistics and other public data. However, these tools differ in the publication systems and data models they target. Table 1 summarizes the main distinctions. Provider-specific packages, such as **eurostat** (Lahti et al., 2017), **WDI** (Arel-Bundock, 2026), and **osmar** (Eugster and Schlesinger, 2013), provide convenient access to individual statistical sources or data families. Broader data-retrieval

packages, such as **pdfetch** (Reinhart, 2024), **webchem** (Szöcs et al., 2020), and **intsvy** (Caro and Biecek, 2017), support broader data collection areas. The **rsdmx** package supports the Statistical Data and Metadata eXchange (SDMX) standard, which is an important alternative standard for statistical data exchange (Blondel et al., 2025; Willekens, 2023). The closest software relatives are direct PxWeb API clients. In R, **PxWebApiData** reads PxWeb data by API and supports both PxWeb API version 1 and 2 (Langsrud et al., 2026). It has been developed in the Statistics Norway software ecosystem for specialized PxWeb data-access workflows, and its documented backend options can use **pxweb** for API capture and data-frame conversion. The **pxweb** package predates these newer clients, with a first CRAN release in 2014, and is positioned as general R infrastructure for PxWeb APIs. In Python, pxwebpy targets PxWeb API version 2 and plays a similar role as **pxweb** does for R (Furne, 2026).

In this landscape, **pxweb** occupies the PxWeb API niche. It provides a common interface for (1) discovering databases and tables, (2) inspecting metadata, (3) constructing valid queries, (4) respecting provider limits, (5) retrieving data, and (6) retaining comments and citation information across statistical organizations that publish through PxWeb. This makes **pxweb** complementary to existing tools. The contribution of **pxweb** is to support the common case where official statistics are distributed through PxWeb APIs and where users need a general, metadata-aware, and reproducible interface from R.

Software type	Packages	Relation to **pxweb**
Provider-specific data clients	**eurostat**, **WDI**, **osmar**	Focus on one data provider or data family
Generic data retrieval tools	**pdfetch**, **webchem**, **intsvy**	Retrieve broad data sources but do not target PxWeb APIs
PX/PC-Axis file tools	**pxR**	Work with local `.px` files
SDMX clients	**rsdmx**	Access data via the alternative SDMX standard.
Direct PxWeb API clients	**PxWebApiData**, pxwebpy	Address PxWeb APIs directly; **PxWebApiData** provides specialized R workflows and can use **pxweb** through optional backend settings and wrappers, while pxwebpy serves Python
Downstream domain packages	**geofi** (Kainu et al., 2026)	Specialized workflows on top of official-data access

Table 1: Related software to **pxweb**.

2 The pxweb package

The **pxweb** package is designed as a generic interface. Its core design goal is to provide a stable R workflow across PxWeb providers while preserving provider-specific metadata where it affects query construction, codelists, comments, and citation. The **pxweb** package turns API workflows into R operations. The package is licensed under the BSD-2-Clause (aka FreeBSD) license, consistent with common practice for research software (Morin et al., 2012).

In the forthcoming version 1.0.0 release described here, the package interface has 39 exported functions, but the main workflow are built from a smaller set of recommended functions. The central workflow is built around the `pxweb_get()` function, which retrieves hierarchy levels, table metadata, or table data depending on the supplied PxWeb version, URL and query. The helper `pxweb_get_data()` wraps the same workflow and returns a `data.frame`. Additional methods, such as `as.data.frame()`, integrate the outputs directly with **tidyverse** workflows (Wickham et al., 2019), allowing users to move from data retrieval to tidy data analysis (Wickham, 2014). Query construction is handled by `pxweb_query()` together with query construction helper functions `pxweb_all()`,

`pxweb_latest()`, and `pxweb_aggregation()`. Table 2 summarizes the main user-facing functions and methods by their role in the recommended workflow.

Workflow part	Recommended functions and methods
Search and discovery	`pxweb_search()`, `pxweb_interactive()`, `pxweb_api_catalogue()`
Metadata	`pxweb_get()`, `pxweb_codelists()`
Query	`pxweb_query()`, `pxweb_validate_query_with_metadata()`, `pxweb_all()`, `pxweb_latest()`, `pxweb_aggregation()`, `pxweb_valueset()`
Retrieval	`pxweb_get()`, `pxweb_get_data()`
Conversion	`as.data.frame()`
Comments and citation	`pxweb_data_comments()`, `pxweb_cite()`

Table 2: Recommended **pxweb** functions and methods.

Internally, the package represents API URLs, metadata, data, comments, and query objects with S3 classes. This allows common operations to behave consistently across API versions. The package uses **httr** for HTTP requests and **jsonlite** for JSON parsing (Wickham, 2026; Ooms, 2014). Table 3 summarizes the main object classes used in ordinary workflows.

Class	Role and common methods
`pxweb`	Represents an API endpoint created from a URL; printed with `print()`
`pxweb_metadata`	Stores table metadata and selected version 2 metadata; used by `pxweb_metadata_dim()`, `pxweb_codelists()`, and `pxweb_validate_query_with_metadata()`
`pxweb_query`	Represents selections to send to the API; created with `pxweb_query()`, converted to JSON with `pxweb_query_as_json()`, and converted back to executable code with `pxweb_query_as_rcode()`
`pxweb_data`, `pxweb_data_v2`	Represent retrieved table data for API versions 1 and 2; created by `pxweb_get()`, converted with `as.data.frame()` or `as.matrix()`, inspected with `pxweb_data_dim()` and `pxweb_data_colnames()`, and cited with `pxweb_cite()`
`pxweb_data_comments`	Collects dataset, dimension, value, and observation comments from `pxweb_data_comments()`; printed or converted to a `data.frame` for inspection and reporting

Table 3: Main S3 object classes used by **pxweb** in ordinary data-access workflows.

The bundled API catalogue is a practical layer on top of the generic API client. It stores provider URL templates, available languages, version paths, aliases, and citation metadata. Users can rely on the catalogue for common providers or pass explicit PxWeb URLs for APIs that are not already bundled with the package. The current catalogue contains 46 entries covering national, regional, and international statistical resources; the complete list is given in the appendix (Table 4). Since PxWeb installations are maintained by the individual data providers, the catalogue is a tested convenience layer rather than a guarantee that every endpoint will remain permanently available. Hence, users can pass explicit PxWeb URLs when catalogue entries change or are missing. At the time of writing, the bundled catalogue entries point mainly to version 1 API paths. Version 2 APIs can also be used by passing explicit PxWeb URLs.

A central implementation detail is that user-supplied URLs are first converted into `pxweb` objects. The constructor (1) parses the URL, (2) detects whether the endpoint belongs to PxWeb API version 1 or 2, (3) retrieves provider configuration metadata, and (4) stores call history in a temporary cache. Provider configuration is normalized into a common internal representation with the number of calls allowed per time window, the length of that time window, and the maximum number of downloadable values or cells. This lets the same retrieval workflow use version 1 `?config` responses and version 2 `/config` responses through a shared interface.

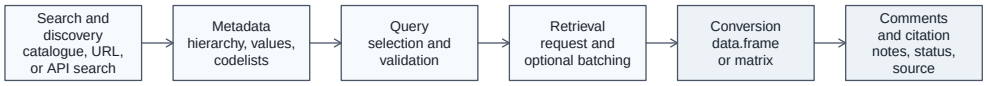


Figure 1: Main **pxweb** workflow: search and discovery, metadata, query construction, retrieval, conversion, comments, and citation.

Support for PxWeb API version 2 is expressed in the same top-level workflow as version 1. Version 2 tables use separate metadata and data endpoints and return JSON-stat2 by default(Statistics Sweden and Statistics Norway, 2026a). The **pxweb** package (1) detects v2 URLs, (2) builds the corresponding data endpoint, (3) constructs the v2 query body, and (4) adds query parameters for codelists and output values when needed. This keeps the user workflow similar across API versions, even though the underlying endpoints and request details differ.

For data requests, `pxweb_get()` first obtains table metadata, then adds mandatory variables (if needed), validates the query against the metadata, and computes any required batches before making data requests. Version 1 tables are queried with the table URL and a JSON request body, whereas version 2 tables use the corresponding `/tables/{id}/data` endpoint with version-specific query parameters. The returned pieces are parsed into package-specific S3 objects.

Large queries of statistical tables can exceed provider-side limits. This makes it hard for the individual user who needs to download the data in multiple batches by partitioning the original query in smaller pieces, download them, and then combine them into one dataset. The `pxweb` package automatically solves this for the user with an in-built batching procedure. This procedure can be viewed as a greedy constrained multidimensional query-partitioning heuristic. Given the provider's maximum downloadable size, the algorithm first identifies splittable dimensions from the table metadata. Time variables can be split, and so can variables marked as eliminable in the provider metadata, meaning that the API allows them to be omitted or partitioned across requests. Content variables and top or bottom selections are kept intact because they affect the meaning or ordering of the returned values. The procedure then evaluates candidate orderings of the splittable dimensions and, for each ordering, greedily chooses the largest dimension chunks that keep each subquery below the provider limit. The ordering with the smallest resulting number of batches is then used to download the query. For the typical case of fewer than seven splittable dimensions, all orderings are evaluated. Otherwise, a random sample of orderings is used to keep computation bounded.

The `pxweb` package follows common software engineering practices, including version control, automated tests, API fixtures, continuous integration, and test workflows for selected APIs (Perez-Riverol et al., 2016). The default local test suite currently uses mocked and offline-safe tests and live contnous integration (CI) tests at GitHub.

2.1 Functionality

The package supports the full workflow shown in Figure 1: search and discovery, metadata inspection, query construction, retrieval, conversion, comments, and citation.

For released versions, installation from CRAN follows the standard procedure in R.

```
install.packages("pxweb")
```

```
library("pxweb")
```

Search and discovery start either from the bundled API catalogue, for PxWeb API version 1, from an explicit URL, or, for PxWeb API version 2, from table search. The following example searches Statistics Sweden's PxWeb API version 2 for a population table and stores its metadata URL.

```
search_results <- pxweb_search(
  query = "Population by region marital status age sex",
  api_url = "https://statistikdatabasen.scb.se/api/v2",
  lang = "en",
  page_size = 5
)

# Select the population table used in the examples
url <- search_results$metadata_url[search_results$id == "TAB638"][1]
```

Metadata are retrieved with `pxweb_get()` when no query is supplied. The returned metadata object describes the table dimensions, values, labels, and version-specific metadata needed for query validation and data retrieval. Version 2 APIs may also expose aggregation codelists and value sets, which can be inspected with `pxweb_codelists()`.

```
meta <- pxweb_get(url)
age_codelists <- pxweb_codelists(meta, variable = "Alder")
age_codelists[, c("id", "type", "label")]
```

```
              id        type                          label
1 agg_Ålder10årJ Aggregation             ´10-year intervals
2   agg_Ålder5år Aggregation               5-year intervals
3   vs_Ålder1årA    Valueset        Age, 1 year age classes
4   vs_ÅlderTotA    Valueset Age, total, all presented ages
```

The codelist labels are provider-supplied. In this example the English metadata endpoint returns English labels, while the `id` values remain provider identifiers that are passed unchanged to query helpers. The five-year age aggregation can therefore be selected from the metadata and stored as an object before query construction.

Queries can be written as named R lists and converted to `pxweb_query` objects when needed. For codelists, the returned `id` column is passed to aggregation or value-set helpers. Query helper functions express common selection patterns such as "all values", "the latest time period", top or bottom selections, aggregations, and value sets. Queries are validated against the metadata of the table before calling the PxWeb API.

```
age_5_year_id <- age_codelists$id[age_codelists$label == "5-year intervals"]

query <- list(
  Region = "00",
  Civilstand = "OG",
  Alder = pxweb_aggregation(age_5_year_id),
  Kon = c("1", "2"),
  ContentsCode = "BE0101N1",
  Tid = pxweb_latest()
)

px_query <- pxweb_query(query)
```

Retrieval is handled by `pxweb_get()`. Users who want a `data.frame` directly can instead use `pxweb_get_data()`. The same top-level workflow handles version 1 and version 2 URLs,

constructs the appropriate request body, and splits large supported queries into batches when needed.

```
px_data <- pxweb_get(url, query = query)
```

Conversion methods turn PxWeb responses into standard R objects. Users can choose whether column names and variable values use provider `label` or `code`.

```
px_df <- as.data.frame(
  px_data,
  column.name.type = "code",
  variable.value.type = "code"
)
```

Finally, PxWeb responses may include comments, notes, or status information. The **pxweb** package exposes these through `pxweb_data_comments()`, with support for both version 1 comments and version 2 dataset notes, dimension notes, value notes, and observation status comments. Citation support is provided by `pxweb_cite()`, which prints citation information for the retrieved data source and the **pxweb** package itself.

```
comments <- pxweb_data_comments(px_data)
pxweb_cite(px_data)
```

3 Case study

We illustrate the package with a Statistics Sweden population table that is both familiar official statistics and technically useful for showing large queries. The example uses Statistics Sweden's PxWeb API version 2 table search to locate table `TAB638`, "Population by region, marital status, age and sex. Year 1968–2024". Exact results can change if Statistics Sweden changes table metadata, values, or provider-side configuration. The replication materials include cached inputs from the original retrieval and use them by default.

```
library("pxweb")

search_results <- pxweb_search(
  query = "Population by region marital status age sex",
  api_url = "https://statistikdatabasen.scb.se/api/v2",
  lang = "en",
  page_size = 5
)

metadata_url <- search_results$metadata_url[search_results$id == "TAB638"]
```

We use a large but manageable subset of the table: all 21 Swedish counties, all marital-status categories, single-year ages from 0 to 100+, both sexes, the population count content variable, and all years from 1968 to 2024. These codes can all be accessed with the metadata object. The marital-status code with a non-ASCII provider character is written with a Unicode escape in the source below.

```
px_metadata <- pxweb_get(metadata_url)
px_metadata

PXWEB METADATA
Population by region, marital status, age and sex.  Year 1968-2024
variables:
```

```
 [[1]] Region: region
 [[2]] Civilstand: marital status
 [[3]] Alder: age
 [[4]] Kon: sex
 [[5]] ContentsCode: observations
 [[6]] Tid: year
```

The metadata object contains the labels returned by the provider. With `lang = "en"` in the search call, the selected metadata URL also requests English labels.

```
years <- as.character(1968:2024)
ages <- c(as.character(0:99), "100+")
county_codes <- c(
  "01", "03", "04", "05", "06", "07", "08", "09", "10", "12", "13",
  "14", "17", "18", "19", "20", "21", "22", "23", "24", "25"
)

query <- list(
  Region = county_codes,
  Civilstand = c("OG", "G", "SK", "\u00c4NKL"),
  Alder = ages,
  Kon = c("1", "2"),
  ContentsCode = "BE0101N1",
  Tid = years
)

pxq <- pxweb_query(query)
pxq
```

The query printout below is abbreviated.

```
PXWEB QUERY
query:
 [[1]] Region (item):
   01, 03, 04, 05, 06, 07, ...
 [[2]] Civilstand (item):
   OG, G, SK, ÄNKL
 [[3]] Alder (item):
   0, 1, 2, 3, 4, 5, 6, 7, ...
 [[4]] Kon (item):
   1, 2
 [[5]] ContentsCode (item):
   BE0101N1
 [[6]] Tid (item):
   1968, 1969, 1970, 1971, ...
```

The complete table is much larger than a typical interactive API request. Its dimensions contain 312 region codes, four marital-status categories, 102 age categories, two sex categories, two content variables, and 57 years. Downloading the full table would imply more than 29 million cells. Statistics Sweden's PxWeb API version 2 configuration currently limits a single data response to 150,000 data cells. Our subset contains 967,176 cells, which is also above the single-response limit. Calling `pxweb_get()` validates the query, splits it into seven batches, downloads the pieces, and combines them into one `pxweb_data_v2` object.

```
px_data <- pxweb_get(metadata_url, query = pxq)

Downloading large query (in 7 batches):
|======================================================================| 100%
```

Comments, notes, and observation statuses can be extracted from the downloaded object before conversion. The same call returns an empty comments data frame for responses without comments. The output below shows two abbreviated comments.

```
comments <- pxweb_data_comments(px_data)
comments_df <- as.data.frame(comments, stringsAsFactors = FALSE)
comments_df[, c("comment_type", "comment")]
```

```
    comment_type    comment
1    obs_comment    Heby municipality changed county affiliation from Upps ...
2 column_comment    The period 1968-1998 is reported according to the regi ...
```

The downloaded object can then be converted directly to a standard `data.frame` for analysis.

```
population <- as.data.frame(
  px_data,
  column.name.type = "code",
  variable.value.type = "code"
)

dim(population)
```

```
[1] 967176      7
```

The downloaded data object can be cited as follows. Below is the data citation from the archived retrieval used in the replication materials.

```
pxweb_cite(px_data)
```

```
Statistics Sweden (2026). "Population by region, marital status, age,
sex and year." [Data accessed 2026-09-01 14:21:25.642737 using pxweb R
package 1.0.0],
<https://statistikdatabasen.scb.se/api/v2/tables/TAB638/data>.
```

Finally, the data are aggregated and plotted. Figure 2 summarizes the downloaded data. The first two panels aggregate counties, marital statuses, ages, and sexes to show the growth of the population and the increasing share of the population aged 65 years or older. The lower panels compare the age and sex structure in 1968 and 2024. The example demonstrates the main package workflow: a PxWeb API version 2 table is found through search, queried programmatically, split because it exceeds the API cell limit, inspected for comments, cited, and transformed into standard R objects for analysis and visualization.

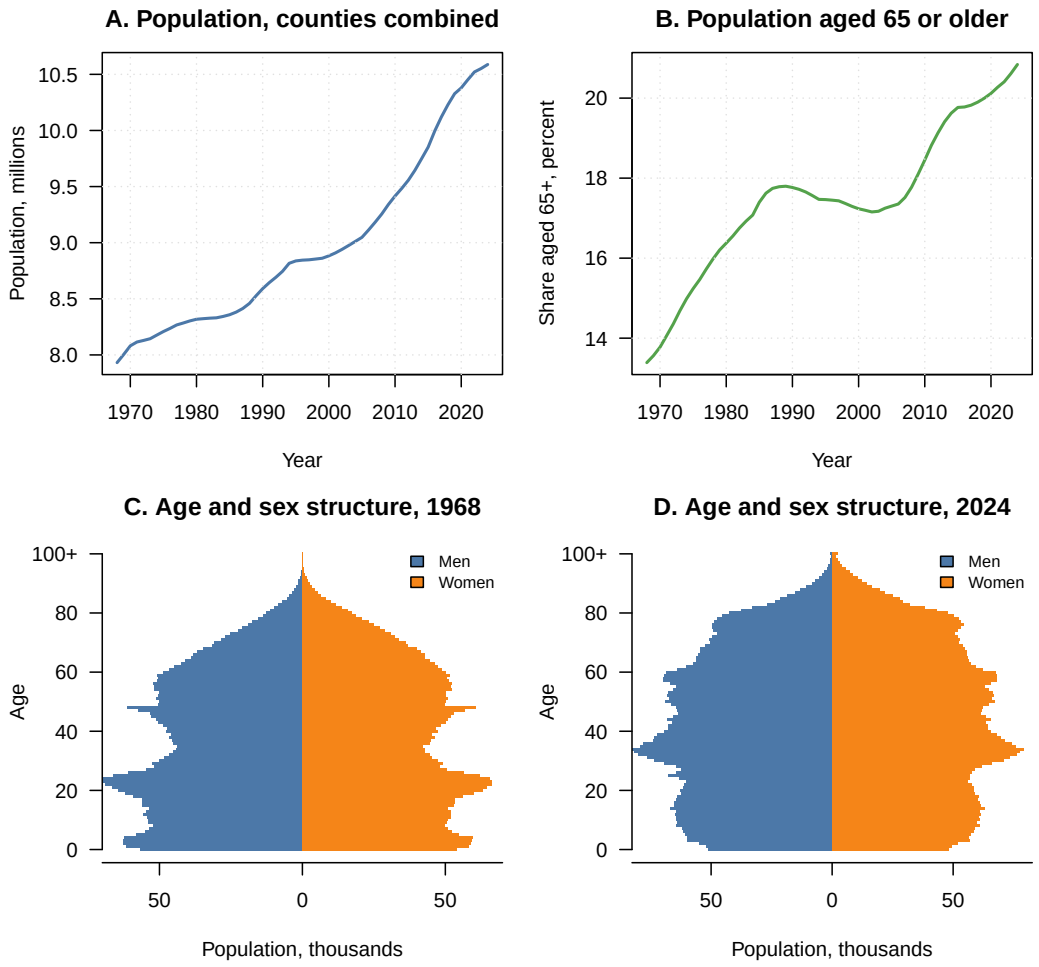


Figure 2: Population development in Sweden based on Statistics Sweden table `TAB638`. The data are downloaded at county level for all years from 1968 to 2024, all single-year age groups from 0 to 100+, both sexes, and all marital-status categories. The request contains 967,176 cells and is split by **pxweb** into seven API calls before being combined for analysis.

4 Discussion

The **pxweb** package provides access to official statistics shared through PxWeb APIs. It supports the full workflow from API discovery and metadata inspection to query construction, retrieval, conversion, comments, codelists, and citation. In doing so, it helps users move from interactive table browsing to scripted, transparent, and reproducible statistical workflows following the tidy data principles (Wickham et al., 2019; Wickham, 2014).

The package has matured through active development since its first CRAN release in 2014 and now include support for PxWeb API version 2. As an indication of use at the time of writing, CRAN download logs report 116,164 downloads between January 2014 and June 2026 (CRAN logs, 2026). The package has also been cited in research using official-statistics data, for instance in work on electronic gambling machines and socioeconomic status (Raisamo et al., 2019), income-inequality data construction (Solt, 2020), trends in youth violence and mortality (Tanskanen et al., 2026; Hakkarainen et al., 2026), radiation-dose modeling after the Chernobyl nuclear power plant accident (Tondel et al., 2025), and programmatic access to open statistical data for population studies (Willekens, 2023).

However, PxWeb installations differ across providers, with different provider-side query limits and rate limits, and APIs can change independently of the **pxweb** package. These issues motivate the package's emphasis on provider config handling, metadata validation, and request splitting. Together this creates a flexibility to handle many different types of PxWeb APIs-

More broadly, **pxweb** contributes to the open data science ecosystem (Lahti, 2018) by making official statistics easier to retrieve, document, and reuse in R. The package does not replace provider documentation, but it gives researchers, public-sector analysts, educators, and civic users a common programmatic interface to a widely used dissemination infrastructure.

5 Appendix: Bundled API catalogue

Entry	Provider	Language
web.dzs.hr	Croatian Bureau of Statistics	en
andmed.stat.ee	Estonia – official statistics	en, et
statistika.tai.ee	Estonian Health Statistics and Health Research Database	en
tilastot.etk.fi	Finnish Centre for Pensions	en
tilastot.tela.fi	Finnish Pension Alliance statistics	en
trafi2.stat.fi	Finnish Transport Safety Agency	en, sv, fi
px.web.ined.fr	Generations and Gender Contextual Database	en
pc-axis.geostat.ge	Geostat Statistics Database	en
stat.hel.fi	Helsingin seudun aluesarjat -tilastotietokanta	fi
pxweb.irena.org	International Renewable Energy Agency	en
www6.poderjudicial.es	Judicial statistics, Spain	es
data.stat.gov.lv	Latvia – official statistics	en, lv
statistika.spkc.gov.lv	Latvian Health Statistics Database	en
statistik.linkoping.se	Linkoping municipality in Sweden	sv
statdb.luke.fi	LUKE Natural Resources Institute Finland	en, fi, sv
pxweb.nhwstat.org	Nordic Health and Welfare Statistics	en
pxweb.nordicstatistics.org	Nordic Statistics Database	en
openstat.psa.gov.ph	Philippine Statistics Authority OpenSTAT	en
www.grande-region.lu	Portail statistique de la Grande Region	fr, de
pxweb.stat.si	SiStat Database	sl
bank.stat.gl	Statbank Greenland	en, kl, da
makstat.stat.gov.mk	State Statistical Office of the Republic of Macedonia	en, mk
pxweb.asub.ax	Statistics Aland	en, sv
statbank.hagstova.fo	Statistics Faroe Islands	en, fo
statfin.stat.fi	Statistics Finland	en, fi, sv
px.hagstofa.is	Statistics Iceland	en, is
askdata.rks-gov.net	Statistics Kosovo	en
etab.llv.li	Statistics Liechtenstein	en
statbank.statistica.md	Statistics Moldova	en, ro
api.scb.se	Statistics Sweden	en, sv
www.pxweb.bfs.admin.ch	Statistics Switzerland	en, de, fr
m02-http-pxwebb.login.sundsvall.se	Sundsvall municipality in Sweden	sv
statistik.tillvaxtanalys.se	Swedish Agency for Growth Policy Analysis	sv
statistik.csn.se	Swedish Board of Student Finance	sv
pxexternal.energimyndigheten.se	Swedish Energy Agency	en
skogsstatistik.slu.se	Swedish University of Agricultural Sciences forest statistics	en
fohm-app.folkhalsomyndigheten.se	The Public Health Agency of Sweden	sv
statistik.sjv.se	The Swedish Agricultural Agency	sv
pxweb.skogsstyrelsen.se	The Swedish Forest Agency	en
statistik.konj.se	The Swedish National Institute of Economic Research	en, sv
prognos.konj.se	Swedish National Institute of Economic Research, forecast database	sv
w3.unece.org	United Nations Economic Commission for Europe	en
statistik.vasteras.se	Vasteras municipality in Sweden	sv
pxweb2022.vgregion.se	Vastra Gotaland Region in Sweden	sv
vero2.stat.fi	Verohallinto – Finnish Tax Administration	en, fi, sv
visitfinland.stat.fi	Visit Finland (Rudolf service)	fi

Table 4: Bundled PxWeb API catalogue entries in **pxweb**. The table lists the catalogue key, provider description, and available languages.

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References

V. Arel-Bundock. *WDI: World Development Indicators and Other World Bank Data*, 2026. URL https://doi.org/10.32614/CRAN.package.WDI. R package version 2.8.0. [p1, 2]

C. J. G. Bellosta, F. J. Viciana, O. P. Lamigueiro, and E. T. Manzanera. *pxR: PC-Axis with R*, 2025. URL https://doi.org/10.32614/CRAN.package.pxR. R package version 0.42.8. [p2]

E. Blondel, M. Stigler, and E. Persson. *rsdmx: Tools for Reading SDMX Data and Metadata*, 2025. URL https://doi.org/10.32614/CRAN.package.rsdmx. R package version 0.6-5. [p1, 3]

C. Boettiger, S. Chamberlain, E. Hart, and K. Ram. Building software, building community: Lessons from the rOpenSci project. *Journal of Open Research Software*, 3(1):e8, November 2015. URL https://doi.org/10.5334/jors.bu. [p1]

D. H. Caro and P. Biecek. intsvy: An R package for analyzing international large-scale assessment data. *Journal of Statistical Software*, 81(7):1–44, 2017. URL https://doi.org/10.18637/jss.v081.i07. [p1, 3]

CRAN logs. Total downloads for the pxweb package, 2026. URL https://cranlogs.r-pkg.org/downloads/total/2014-01-01:2026-06-30/pxweb. Accessed 2026-06-30. [p11]

M. J. A. Eugster and T. Schlesinger. osmar: OpenStreetMap and R. *The R Journal*, 5(1):53–63, June 2013. URL https://doi.org/10.32614/RJ-2013-005. [p1, 2]

S. Furne. *pxwebpy: Client Library for the PxWeb API to Easily Load Data into a DataFrame*, 2026. URL https://pypi.org/project/pxwebpy/. Python package version 0.6.0. [p3]

C. Gandrud. *Reproducible Research with R and R Studio*. Chapman & Hall/CRC, July 2013. [p1]

R. M. Groves and L. Lyberg. Total survey error: Past, present, and future. *Public Opinion Quarterly*, 74(5):849–879, 01 2010. URL https://doi.org/10.1093/poq/nfq065. [p1]

P. Hakkarainen, P. Kantanen, K. Kataja, J. Kivivuori, L. Lahti, T. Pitkänen, S. Rönkä, and A. Seppänen. Näkökulmia nuorten itsemurha-, huume- ja väkivaltakuolemiin: Sosiaalinen kuolemansyytutkimus ja epätoivokuolemat. *Yhteiskuntapolitiikka*, 91(2):149–157, 2026. [p11]

M. Kainu, J. Lehtomäki, J. Parkkinen, J. Miettinen, P. Kantanen, S. Vesanen, and L. Lahti. *geofi: Access Finnish Geospatial Data*, 2026. URL https://doi.org/10.32614/CRAN.package.geofi. R package version 1.2.1. [p3]

L. Lahti. Open data science. In *Advances in Intelligent Data Analysis XVII*, volume 11191 of *Lecture Notes in Computer Science*, pages 31–39. Springer-Verlag, 2018. URL https://doi.org/10.1007/978-3-030-01768-2_3. [p11]

L. Lahti, J. Parkkinen, J. Lehtomäki, and M. Kainu. rOpenGov: Open source ecosystem for computational social sciences and digital humanities. Int'l Conf. on Machine Learning - ICML/MLOSS Open Source Software workshop, December 2013. URL http://ropengov.github.io. [p2]

L. Lahti, J. Huovari, M. Kainu, and P. Biecek. Retrieval and analysis of Eurostat open data with the eurostat package. *The R Journal*, 9(1):385–392, 2017. URL https://doi.org/10.32614/RJ-2017-019. [p1, 2]

Ø. Langsrud, J. Bruusgaard, S. Bjørkholt, and S. Jentoft. *PxWebApiData: PX-Web Data by API*, 2026. URL https://doi.org/10.32614/CRAN.package.PxWebApiData. R package version 1.9.0. [p3]

P. Likidis. *PX-File Format Specification: AXIS-VERSION 2013*. Statistics Sweden, 2013. URL https://www.scb.se/globalassets/vara-tjanster/px-programmen/px-file_format_specification_2013.pdf. Accessed 2026-06-30. [p2]

M. Magnusson, M. Kainu, J. Huovari, and L. Lahti. *pxweb: R Interface to PXWEB APIs*, 2026. URL https://github.com/rOpenGov/pxweb/. R package version 1.0.0. [p1]

A. Morin, J. Urban, and P. Sliz. A quick guide to software licensing for the scientist-programmer. *PLoS Computational Biology*, 8(7):e1002598, 2012. URL https://doi.org/10.1371/journal.pcbi.1002598. [p3]

J. Ooms. The jsonlite package: A practical and consistent mapping between JSON data and R objects. *arXiv:1403.2805 [stat.CO]*, 2014. URL http://arxiv.org/abs/1403.2805. [p4]

M. A. Parsons, R. E. Duerr, and M. B. Jones. The history and future of data citation in practice. *Data Science Journal*, 18:52, Nov 2019. URL https://doi.org/10.5334/dsj-2019-052. [p1]

Y. Perez-Riverol, L. Gatto, R. Wang, T. Sachsenberg, J. Uszkoreit, F. da Veiga Leprevost, C. Fufezan, T. Ternent, S. J. Eglen, D. S. Katz, T. J. Pollard, A. Konovalov, R. M. Flight, K. Blin, and J. A. Vizcaíno. Ten simple rules for taking advantage of git and GitHub. *PLoS Computational Biology*, 12(7):e1004947, July 2016. URL https://doi.org/10.1371/journal.pcbi.1004947. [p5]

S. Raisamo, A. Toikka, J. Selin, and M. Heiskanen. The density of electronic gambling machines and area-level socioeconomic status in finland: A country with a legal monopoly on gambling and a decentralised system of EGMs. *BMC Public Health*, 19(1):1198, 2019. URL https://doi.org/10.1186/s12889-019-7535-1. [p11]

A. Reinhart. *pdfetch: Fetch Economic and Financial Time Series Data from Public Sources*, 2024. URL https://doi.org/10.32614/CRAN.package.pdfetch. R package version 0.3.3. [p1, 3]

F. Solt. Measuring income inequality across countries and over time: The standardized world income inequality database. *Social Science Quarterly*, 101(3):1183–1199, may 2020. URL https://doi.org/10.1111/ssqu.12795. [p11]

Statistics Finland. Px suite, 2026. URL https://stat.fi/en/services/statistical-data-services/statistical-databases/px-suite. Accessed 2026-06-30. [p2]

Statistics Norway. Public apis, 2026. URL https://www.ssb.no/en/api/pxwebapi. Accessed 2026-06-30. [p2]

Statistics Sweden. Statistical programs for px files, 2026. URL https://www.scb.se/en/services/statistical-programs-for-px-files/. Accessed 2026-06-30. [p2]

Statistics Sweden and Statistics Norway. What is PxWeb?, 2026a. URL https://www.pxtools.net/PxWeb2/project/introduction/. PxTools documentation. Accessed 2026-06-30. [p1, 2, 5]

Statistics Sweden and Statistics Norway. PxWeb 2 plans, 2026b. URL https://www.pxtools.net/PxWeb2/project/plans/. PxTools documentation. Accessed 2026-08-30. [p2]

Statistics Sweden and Statistics Norway. The project new interface for PxWeb, 2026c. URL https://www.pxtools.net/PxWeb2/project/process/. PxTools documentation. Accessed 2026-08-30. [p2]

Statistics Sweden and Statistics Norway. PxWebApi 2 user guide, 2026d. URL https://www.pxtools.net/PxWebApi/documentation/user-guide/. PxTools documentation. Accessed 2026-08-30. [p2]

E. Szöcs, T. Stirling, E. R. Scott, A. Scharmüller, and R. B. Schäfer. webchem: An R package to retrieve chemical information from the Web. *Journal of Statistical Software*, 93(13):1–17, 2020. URL https://doi.org/10.18637/jss.v093.i13. [p1, 3]

M. Tanskanen, A. Beuker, K. Suonpää, K. Haapakangas, and J. Kivivuori. Nuorten vakavan väkivallan kehitystrendit eri osoittimien valossa: Toistettava menetelmä avoimien tilastotietokantojen analyysiin. *Kriminologia*, 7(1):1–19, 2026. URL https://doi.org/10.54332/krim.161322. [p11]

M. Tondel, K. Gabrysch, C. Rääf, and M. Isaksson. A model for estimating radiation doses and population cancer risk in sweden after the Chernobyl nuclear power plant accident in 1986. Technical report, Uppsala University, 2025. URL https://doi.org/10.33063/diva-544175. [p11]

United Nations General Assembly. Fundamental principles of official statistics. Resolution A/RES/68/261, 2014. URL https://docs.un.org/en/A/RES/68/261. Adopted by the General Assembly on 29 January 2014. [p1]

H. Wickham. Tidy data. *Journal of Statistical Software*, 59(10):1–23, 2014. URL https://doi.org/10.18637/jss.v059.i10. [p3, 11]

H. Wickham. *httr: Tools for Working with URLs and HTTP*, 2026. URL https://doi.org/10.32614/CRAN.package.httr. R package version 1.4.8. [p4]

H. Wickham, M. Averick, J. Bryan, W. Chang, L. D. McGowan, R. François, G. Grolemund, A. Hayes, L. Henry, J. Hester, M. Kuhn, T. L. Pedersen, E. Miller, S. M. Bache, K. Müller, J. Ooms, D. Robinson, D. P. Seidel, V. Spinu, K. Takahashi, D. Vaughan, C. Wilke, K. Woo, and H. Yutani. Welcome to the tidyverse. *Journal of Open Source Software*, 4(43):1686, 2019. URL https://doi.org/10.21105/joss.01686. [p3, 11]

F. Willekens. Programmatic access to open statistical data for population studies: The SDMX standard. *Demographic Research*, 49(40):1117–1162, 2023. URL https://doi.org/10.4054/DemRes.2023.49.40. [p3, 11]

World Bank. World development indicators. https://datatopics.worldbank.org/world-development-indicators/, 2026. Accessed 2026-06-30. [p1]

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